

Method	Outcome information	Profile information	Diagnostic role	Main limitation
Weighted marginal prior	Training-fold category shares	None	Base-rate calibration without personalization	Cannot recover subgroup differences
Multinomial logit	Training-fold individual outcomes	Same features shown to the LLM	Transparent conditional benchmark	Additive log-odds predictor
Histogram gradient boosting	Training-fold individual outcomes	Same features shown to the LLM	Flexible information-ceiling benchmark	Task-specific and outcome-informed
Qwen3-8B	No CGSS outcomes	Persona text from the same features	Zero-shot test of implicit base rates and conditioning	Model-, prompt-, and decoding-specific